CHE 321 Chemical Reaction Engineering
EXAM 1, 100 points
<u>Closed Book, Closed Note (40 Points)</u>

Name KEY

1. (7 points) In a flow system with gas-phase reaction, how do you express the concentration of a component (C_j) as a function of T, P, T_0, P_0, C_{T0}, F_T and F_j? Hint: Think of Ideal Gas Law.

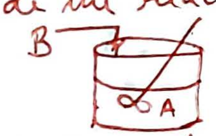
$$C_j = C_{T0} \frac{F_j}{F_T} \cdot \frac{P}{P_0} \cdot \frac{T_0}{T}$$

2. (10 points) Two parallel reactions with two reactants, A and B, are taking place. The rate of the reaction that produces the desired product, A + B → D, is $-r_A = kC_A^2C_B$. The undesired product is produced via A + B → U, $-r_A = kC_AC_B^2$. What reactor setups can provide a better selectivity for the production of D over U. At t=0, $C_A=C_{A0}$, $C_B=C_{B0}$, $C_C=0$, $C_D=0$.

$$S_{D/U} = \frac{kC_A^2C_B}{kC_AC_B^2} = C_A/C_B .$$

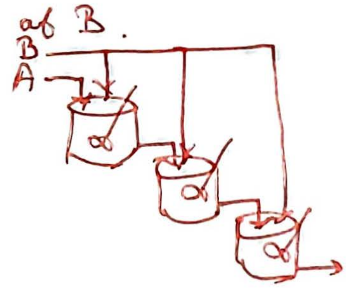
For higher selectivity, we need higher concⁿ of A and lower B.

(a) Semi-Batch Reactor with A inside the reactor & B entering.



(b) PFR w/ side stream for B

(c) Cascaded CSTR w/ side-stream of B.



3. (8 points) What reaction condition/s will result in a higher selectivity of D over U?
A + B → D ($-r_A = k_1 C_A^2 C_B^3$) Activation Energy = 150 eV/molecule
A + B → U ($-r_A = k_2 C_A^2 C_B^3$) Activation Energy = 100 eV/molecule

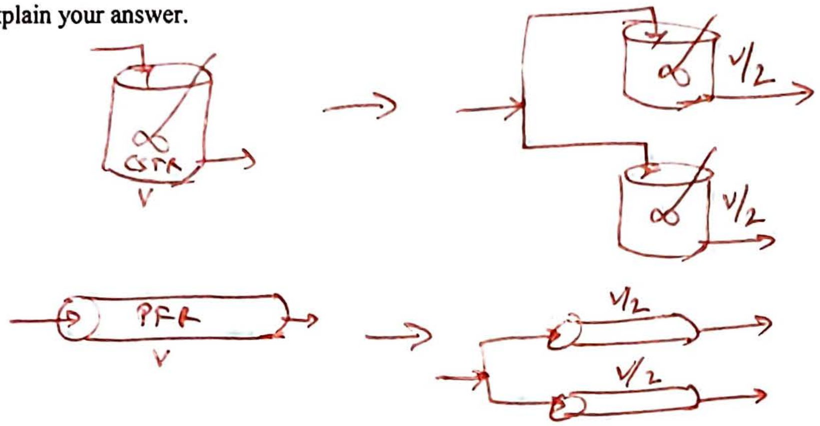
$$S_{D/U} = k_1/k_2 = A_1/A_2 \exp\left[-\frac{(E_D - E_U)}{RT}\right]$$

$E_D > E_U \Rightarrow$ higher T will lead to higher $S_{D/U}$.

Reaction condition → higher T.

4. (8 points) An isothermal reaction is carried out in a reactor of volume V. The same reaction is then carried out in two same type of reactors in parallel, each of volume V/2. The change in conversion is measured. Select one.
The change in conversion:
   a. will be higher if the reactors are CSTR than if they are PFR
   b. will be higher if the reactors are PFR than if they are CSTR
   c. is the same

Explain your answer.



Effective volume = V

Effective volume = V

5. (7 points) Write the mass-balance equation that can be used to study the startup of a CSTR. Write the initial conditions. Do not solve the equation.

$$F_{A0} - F_A + r_A V = \frac{dN_A}{dt}$$

$N_A = C_A V$

$F_A = C_A v$

$F_{A0} = C_{A0} v$

$$v(C_{A0} - C_A) + r_A V = V\frac{d(C_A)}{dt}$$

@ $t = 0$ $\quad$ $C_A = 0$

Open Book/Notes Section (60 Points)

Name ______________________________

The following liquid-phase hydration reaction occurs in a 10,000 L CSTR:

$$A \xrightarrow{H_2O} B + C$$

With a first-order rate constant of 2.5×10^{-3} min^{-1}.

a) **(15 points)** What is the steady-state exit-concentration of A if the feed rate is 25 L/min and the feed concentration $C_{A0} = 0.1$ mol/L?

b) **(30 points)** If the volumetric flowrate suddenly drops to 50% of its original value and is maintained there, what is the conversion of A after 800 minutes, and what is the new steady state conversion? [Hint: This is an unsteady state problem.]

c) **(15 points)** After the steady-state concentration has been reached (in part (b)), the exit stream from the CSTR enters a PFR with volume 5,000 L. What is the exit concentration coming out of the PFR?

(a) $F_{A0} X = (-r_A) V$ $\qquad C_{A0} = 0.1$ mole/L

$\cancel{C_{A0}} v_0 X = k \cancel{C_{A0}} (1-X) V$ $\qquad V = 10,000$ L

$\qquad v_0 = 25$ L/min

$25 X = 2.5 \times 10^{-3} \times 10,000 \times (1-X)$ $\qquad k = 2.5 \times 10^{-3}$ min^{-1}

$X = 1 - X \Rightarrow X = 1/2$

$C_A = C_{A0}(1-X) = 0.1/2 = 0.05$ mole/L

(b) $v \rightarrow 25$ L/min to 12.5 L/min $\qquad t = 800$ min.

C_A @ $t = 0 = 0.05$ mole/L (not 0)

$F_{A0} - F_A + (r_A) V = V \dfrac{dC_A}{dt}$ $\qquad \tau = \dfrac{10000 \text{ L}}{12.5 \text{ L/min}} = 800$ min

$C_{A0} - C_A - k\tau C_A = \tau \dfrac{dC_A}{dt}$ $\qquad C_{A0} = 0.1$ mole/L

$\qquad k\tau = 2.5 \times 10^{-3} \times 800 = 2$

$\left(\dfrac{dC_A}{dt} + \left(\dfrac{1+k\tau}{\tau}\right) C_A = \dfrac{C_{A0}}{\tau}\right) \exp\left[\dfrac{1+k\tau}{\tau} t\right]$

$\dfrac{d}{dt}\left(C_A \exp\left[\dfrac{(1+k\tau)}{\tau} t\right]\right) = \dfrac{C_{A0}}{\tau} \exp\left[\dfrac{(1+k\tau)}{\tau} t\right]$

$C_A \exp\left(\dfrac{(1+k\tau)}{\tau} t\right) \Big|_{C_A|_{t=0},\, 0}^{C_A,\, t} = \dfrac{C_{A0}}{\tau \dfrac{(1+k\tau)}{\tau}} \exp\left[\dfrac{(1+k\tau)}{\tau} t\right] \Big|_0^t$

$$C_A \exp\left(\frac{(1+k\tau)}{\tau}t\right) - C_{A,t=0} = \frac{C_{A0}}{1+k\tau}\left(\exp\left(\frac{(1+k\tau)}{\tau}t\right) - 1\right)$$

$$C_A = C_{A,t=0}\exp\left[-\frac{(1+k\tau)}{\tau}t\right] + \frac{C_{A0}}{1+k\tau}\left(1 - \exp\left[-\frac{(1+k\tau)}{\tau}t\right]\right)$$

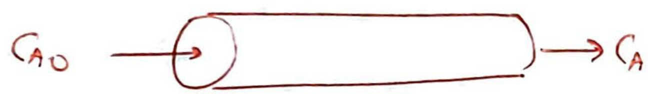
@ $t = 800$ min

$$C_A = 0.05\exp\left[-\frac{3}{800}\times 800\right] + \frac{0.1}{1+2}\left(1 - \exp\left(-\frac{3}{800}\times 800\right)\right)$$

$$= 0.0341 \text{ mole/L}$$

$$C_{A,ss} = \frac{0.1}{3} = 0.033 \text{ mole/L.}$$

(c)



$C_{A0} = 0.033$ mole/L

$V = 5000$ L

$v_0 = 12.5$ L/min

$k = 2.5\times10^{-3}$ /min.

$$v_0\frac{dC_A}{dV} = -kC_A \Rightarrow \ln(C_A/C_{A0}) = -\frac{kV}{v_0}$$

$$C_A = C_{A0}\exp\left[-\frac{kV}{v_0}\right]$$

$$= 0.033\exp\left[-\frac{2.5\times10^{-3}\times 5000}{12.5}\right]$$

$$= 0.0122 \text{ mole/L}$$